

Institute of Actuaries of India

Subject CT1 – Financial Mathematics

March 2017 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

- a) Debentures are part of the loan capital of the company. The term “loan capital” usually refers to long term borrowings rather than short term. The issuing company provides some form of security to holders of the debenture. This is usually in the form of a floating charge against the assets of the company.
- b) Unsecured loan stocks have no explicit assets backing them and holders rank alongside other unsecured creditors. Yields will be higher than on the debentures to reflect the higher risk of default. (2)

ii) Reasons:

- The return on property may be lower, especially in the short term.
- There is less flexibility because of the large unit sizes.
- It may be difficult and expensive to value the property.
- There are large buying and selling expenses.
- The maintenance costs associated with property investment are also high.
- There may be periods when the property is void and no income is received.
- Marketability is poor

(2)

iii)

- a) Government may raise money by floating a loan on stock exchange. The terms of the issue are set out by tender. The investors are invited to nominate the price that they are prepared to pay and the loan is issued to the highest bidders, subject to certain rules of allocation. (2)

b) Advantages:

The government will know the price that the investor will pay at outset so it will know the cost of borrowing money.

It will be administratively less difficult by tender.

Disadvantages:

The investor may be willing to pay more for the bond than the set price and so the money could be borrowed more cheaply if the tender is used

Insufficient investor may be prepared to pay the set price causing only part of the offer to be sold and the Government may not meet its finance requirement (2)

c) Securities with uncertain income include:

- 1) Equities – dividend vary according to the performance of the company issuing stocks and may be zero
- 2) Property – rent payments are subject to review.
- 3) Index-linked bonds – coupon payment and final redemption price may increase in proportion to the increase in the relevant index of inflation (3)

[11 Marks]

Solution 2:

i)

$$a) \quad 1/(1+i^{(2)}/2)^2 = (1-0.06/4)^4$$

$$(1+i^{(2)}/2)^2 = 1.0623193$$

$$i^{(2)} = 6.1377\% \text{ p.a. convertible half-yearly} \quad (2)$$

$$b) \quad (1-d^{(12)}/12)^{12} = (1-0.06/4)^4$$

$$(1-d^{(12)}/12)^{12} = 0.941337$$

$$d^{(12)} = 6.0303\% \text{ convertible monthly} \quad (2)$$

ii) a) Original term of the loan = 365 day

No deferment – settlement date is actual date of repayment 17th July 2005

Outstanding term of the loan = 272 days.

$$\text{The rebate allowed} = (272/365) \times 20000 = 14904.1096$$

$$\text{Sum paid to settle the loan} = (220000 - 14904.1096) = 205095.89 \quad (1.5)$$

b) The loan was repaid after 93 days, the APR obtained is:

$$200000(1+i)^{93/365} = 205095.89$$

$$\Rightarrow i = 10.3\%$$

(1.5)

[7 Marks]

Solution 3:Working in units of Rs 1 crores and assuming i as the rate of interest:

i) The present value of outlay is :

$$2.7 + 0.2 a_{3|} @ i$$

The expected present value of the income is :

$$0.1 v^3 + \bar{a}_{10|} (0.25 v + 0.2 v^2 + 0.1 v^3) @ i$$

Equating the present value of the outlay to the present value of the income and solving for i :

$$2.7 + 0.2 a_{3|} = 0.1 v^3 + \bar{a}_{10|} (0.25 v + 0.2 v^2 + 0.1 v^3) @ i$$

$$\text{This gives } i = 9.3\% \quad (5)$$

ii) Based on the information, the present value of the company's income is :

$$0.1 v^3 + 0.85 \bar{a}_{10|} v^3$$

$$0.1 v^3 + 0.85 \bar{a}_{10|} v^3 - 2.7 - 0.2 a_{3|} @ i = 10\%$$

$$0.1 \times 0.7513 + 0.85 \times 0.7513 \times 6.4469 - 2.70 - 0.2 \times 2.4868$$

$$= 4.19 - 3.19$$

$$= 1.0089$$

Since NPV is positive it is worth to invest in the project.

(3)

[8 Marks]

Solution 4:**i) TWRR**

$$(1+i)^3 = 33/30.50 * (41.05-4.5)/(33+6) * 45.6/41.05 * 47/(45.6-2.50)$$

$$(1+i)^3 = 1.081967 * 0.937179 * 1.11084 * 1.090487$$

$$(1+i)^3 = 1.228313$$

$$(1+i) = 1.070951$$

$$\text{TWRR} = 7.0951\%$$

MWRR

$$30.5*(1+i)^3 + 6*(1+i)^{2.25} + 4.5*(1+i)^{1.5} - 2.5*(1+i)^{0.5} = 47$$

$$\text{At } i = 7\%, \text{ LHS} = 46.74$$

$$\text{At } i = 7.5\%, \text{ LHS} = 47.37$$

By interpolation,

$$\text{MWRR} = 7.206\%$$

(5)

ii) Linked rate of return

Year 2011 – equation of value

$$30.5*(1+i) + 6*(1+i)^{0.25} = 38.50$$

$$i = 6\%, \text{ LHS} = 38.42$$

$$i = 6.5\%, \text{ LHS} = 38.57$$

By interpolation, $i = 6.266\%$

Year 2012 – equation of value

$$38.50*(1+i) + 4.50*(1+i)^{0.50} = 45$$

$$i = 4.5\%, \text{ LHS} = 44.83$$

$$i = 5\%, \text{ LHS} = 45.03$$

By interpolation

$$i = 4.92\%$$

Year 2013 – equation of value

$$45*(1+i) - 2.50*(1+i)^{0.5} = 47$$

$$i = 10\%, \text{ LHS} = 46.88$$

$$i = 10.5\%, \text{ LHS} = 47.097$$

By interpolation

$$i = 10.276\%$$

Linked rate of return

$$(1+i)^3 = (1+0.06266)*(1+0.0492)*(1+0.10276)$$

$$(1+i)^3 = 1.2295$$

$$i = 7.13\%$$

(5)

- iii) TWRR eliminates the effect of the cashflow amount and timing, and therefore gives a fairer basis for assessing the investment performance for the fund. MWRR is sensitive to the amount and timing of the net cashflows.

(1)

[11 Marks]

Solution 5:

- i) Forward rates $f_{1,1}$ and $f_{2,1}$ using the spot rates y_1, y_2 and y_3 ,

$$(1+y_1)^2 = (1+y_1)(1+f_{1,1}) \text{ and}$$

$$(1+y_3)^3 = (1+y_2)^2(1+f_{2,1})$$

$$(1.042)^2 = (1.041)(1+f_{1,1})$$

$$f_{1,1} = 4.30\%$$

and

$$(1.043)^3 = (1.042)^2(1+f_{2,1})$$

$$f_{2,1} = 4.50\%$$

(4)

- ii) Price per 1,000 nominal

$$= 30(v_{4.1\%} + v_{4.2\%}^2 + v_{4.3\%}^3) + 1100 v_{4.3\%}^3$$

$$= 30(0.96061 + 0.92101 + 0.88135) + 1100 \times 0.88135$$

$$= 82.8891 + 969.485$$

$$= 1052.3741$$

(3)

[7 Marks]

Solution 6:

- i) $\int_0^t \delta(t) dt = -\log v(t)$

$$\delta(t) = -(\dot{v}(t)/v(t))$$

$$\delta(t) = (2t + 2\alpha + 1) / ((t + \alpha)(t + \alpha + 1))$$

(3)

- ii) Measure time in years from the start of the given year

Given $\delta(0) = 0.15$ and let

$$\delta(t) = 0.15 + bt + ct^2 \quad (A)$$

Putting $t = \frac{1}{2}$ in the equation (A)

$$0.10 = 0.15 + \frac{1}{2}b + \frac{1}{4}c$$

Putting $t = 1$ in the equation (A)

$$0.08 = 0.15 + b + c$$

Solving the above two equations, $b = -0.13$ and $c = 0.06$

Putting the values of b and c in equation A

$$\delta(t) = 0.15 - 0.13t + 0.06t^2$$

This implies that $\int_0^1 \delta(t) dt = 0.105$

The accumulated amount at the end of year is:

$$1,50,000 \times \exp(0.105) = 1,50,000 \times 1.1107 = \text{Rs}1,66,606.59$$

(5)

[8 Marks]

Solution 7:

- i) The value of the annuity is $\sum_{t=1}^n [(n+1)t - t^2] v^t = (n+1) (Ia)_n - X$
Where $X = \sum_{t=1}^n t^2 v^t$

$$vX = v^2 + 4v^3 + \dots + (n^2 - 2n + 1) v^n + n^2 v^{n+1}$$

$$(1-v)X = 2(Ia)_n - a_n - n^2 v^{n+1}$$

$$X = \frac{2(Ia)_n - a_n - n^2 v^{n+1}}{1-v} \quad (4)$$

ii)

- a) Present value and discounted mean term of the payment received:

$$V_1 = 10000 a_{10} = 61445.67 \text{ at } 10\%$$

$$\text{DMT}_1 = (Ia)_n / a_{10} = 4.7255$$

Present value and discounted mean term of the payment to be made:

$$V_2 = -30000 (v^5 + v^{10}) = -30193.9$$

$$\text{DMT}_2 = (5v^5 + 15v^{10}) / (v^5 + v^{10}) = 6.9153 \quad (4)$$

- b) Discounted mean term of the investor's net cashflow:

$$\text{DMT}_3 = (V_1 \times \text{DMT}_1 + V_2 \times \text{DMT}_2) / (V_1 + V_2) = 2.61 \quad (1)$$

[9 Marks]

Solution 8:

i) $(1+i_t) \sim \text{LN}(\mu, \sigma^2)$

$$\ln(1+i_t) \sim N(\mu, \sigma^2)$$

$$\ln[(1+i_t)^{10}] = \ln(1+i_t) + \ln(1+i_t) + \dots + \ln(1+i_t) \sim N(10\mu, 10\sigma^2)$$

$$(1+i_t)^{10} = \text{LN}(10\mu, 10\sigma^2)$$

$$E(1+i_t) = \exp\left(\mu + \frac{\sigma^2}{2}\right) = 1.06$$

$$\text{Var}(1+i_t) = \exp(2\mu + \sigma^2) * [\exp(\sigma^2) - 1] = 0.08^2$$

$$\frac{0.08^2}{1.06^2} = \exp(\sigma^2) - 1$$

$$\sigma^2 = 0.0056798$$

$$\exp\left(\mu + \frac{0.0056798}{2}\right) = 1.06$$

$$\mu = 0.055429$$

$$\ln(S_{10}) \sim N(10*0.055429, 10*0.0056798)$$

$$S_{10} = \ln(10*0.055429, 10*0.0056798)$$

$$E(S_{10}) = \exp\left(10*0.055429 + 10 * \frac{0.0056798}{2}\right) = 1.790848$$

Thus expected value is Rs 1,79,08,480

(2)

ii) $\Pr(S_{10}) < 1.61172$

$$\Pr(\ln S_{10}) < \ln 1.61172 \text{ where } \ln(S_{10}) \sim N(0.55429, 0.056798)$$

$$\Pr(Z) = \frac{\ln(S_{10}) - 0.55429}{\sqrt{0.056798}} < \frac{\ln 1.61172 - 0.55429}{\sqrt{0.056798}} = -0.32304$$

$$\Pr(Z) < 1 - \Phi(0.32304) \text{ where } Z \sim N(0, 1)$$

$$= 1 - 0.627 = 0.373$$

(Alternate solution for (i) expected value = $1 * 1.06^{10} = 1.790848$ crore)

(6)

[8 Marks]

Solution 9:

The amount of original loan was 1000 a_{25} at 8%

Fifteen payments remain to be made, the t th of which is the $(10+t)$ th overall payment.

The t^{th} remaining payment consist of capital $1000 v^{16-t}$ at 8%

And the interest $1000 (1 - v^{16-t})$ at 8%

After tax the investor receives only 60% of the interest content of each payment.

To obtain a yield of 10% p.a. he should pay

$$\sum_{t=1}^{15} (1.1)^{-t} [1000 v^{16-t} + (1-0.4) 1000 (1 - v^{16-t})]$$

$$= \sum_{t=1}^{15} (1.1)^{-t} (600 + v^{16} \times 400 \times v^{-t})$$

$$= 600 a_{15} \text{ at } 10\% + 1.08^{-16} \times 400 \times \frac{1 - (\frac{1.08}{1.1})^{15}}{1.1 - 1.08}$$

$$= 5968.27$$

[7 Marks]

Solution 10:

- i) The purchase of a long-dated fixed interest swap to pay floating and receive fixed would enable the insurance company to receive a fixed interest rate payments appropriate to immunise the liabilities. The commitment to pay floating would be met by the returns from the investment made by the insurance company.

(2)

- ii) Present value of the dividends is

$$= 20 [1.01 e^{\frac{-0.05}{4}} + (1.01)^2 e^{\frac{-0.05}{2}} + \dots + (1.01)^8 e^{-0.10}]$$

$$+ 20 (1.01)^8 e^{-0.10} [1.015 e^{\frac{-0.05}{4}} + (1.015)^2 e^{\frac{-0.05}{2}} + \dots + (1.015)^4 e^{-0.05}]$$

$$= 20 \times 1.01 x e^{\frac{-0.05}{4}} \left[\frac{1 - (1.01 e^{\frac{-0.05}{4}})^8}{1 - (1.01 e^{\frac{-0.05}{4}})} \right] + 20 \times 1.01^8 x e^{-0.1125} \left[\frac{1 - (1.015 e^{\frac{-0.05}{4}})^4}{1 - (1.015 e^{\frac{-0.05}{4}})} \right]$$

$$= 158.177 + 78.854$$

$$= 237.031$$

So the forward price is

$$(450 - 237.031) e^{0.15} = 247.4$$

(7)

[9 Marks]

Solution 11:

i) Let the loan amount be Rs L

$$L = 180000 \times a_{\overline{15}|i} + 20000 \times (Ia)_{\overline{15}|i} \quad @12\%$$

$$L = 180000 \times 6.8109 + 20000 \times (\ddot{a}_{\overline{15}|i} - 15 \times v^{15}) / i$$

$$L = 12,25,962 + 20,000 \times (7.6282 - 2.74044) / 0.12$$

$$L = 12,25,962 + 20,000 \times 40.7313 = 20,40,588$$

Amount of Loan from the bank = Rs 20,40,588

$$\text{Total cost of house} = 2040588 / 0.8 = \text{Rs } 25,50,735$$

(4)

ii) The loan outstanding at the beginning of the ninth year :

$$L = (180000v + 180000v^2 + \dots + 180000v^7) + (9 \times 20000v + 10 \times 20000v^2 + \dots + 15 \times 20000v^7)$$

$$L(9) = 3,40,000 \times a_{\overline{7}|i} + 20000 \times (Ia)_{\overline{7}|i} \quad @12\%$$

$$L(9) = 3,40,000 \times 4.5638 + 20,000 \times (5.1114 - 7 \times 0.45235) / 0.12$$

$$L(9) = 1551672 + 324158.33 = \text{Rs } 18,75,850.33$$

Loan Schedule:

Year	Loan o/s	Loan Installment	Interest	Capital repayment
9	18,75,850.33	3,60,000	2,25,102.04	1,34,897.96
10	17,40,952.37	3,80,000	2,08,914.28	1,71,085.72
11	15,69,866.65			

(5)

iii) Let the new installment be Rs P

$$15,69,866.65 = P \times a_{\overline{5}|i} \quad @10\%$$

$$15,69,866.65 = P \times 3.790786$$

$$P = 4,14,126.87$$

Extra payment made by Mr Sam if the loan continued with the first bank:

Total Installments less Loan Outstanding at the end of 10th year

$$(400000+420000+440000+460000+480000) - 15,69,866.65$$

$$= 6, 30, 133.35$$

Extra payment made by Mr Sam if the loan transferred to second bank:

Total Installments + 1% processing fees - Loan Outstanding at the end of 10th year

$$= 5 \times 414126.90 + .01 \times 15,69,866.65 - 15,69,866.65$$

$$= 20,70,634.337 + 15,698.665 - 15,69,866.65$$

$$= 5,16,466.35$$

Since the extra payment (interest + processing fee) to second bank is lower than the payments to the first bank,
it is advisable to Mr Sam to transfer his loan to the second bank.

(6)

[15 Marks]
